

Exp No:1.a Analyze the trend of data science job postings over the last decade

Aim:

To analyze the trend of data science job postings over the last decade

Description:

Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation

and matplotlib/seaborn for visualization.

code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
data = {'Year': list(range(2009, 2021)),
```

```
'Job Postings': [100, 350, 400, 650, 800, 1250, 1600, 2150, 2700, 3300, 4200, 4400]}
```

```
df = pd.DataFrame(data)
```

```
plt.plot(df['Year'], df['Job Postings'], marker='o')
```

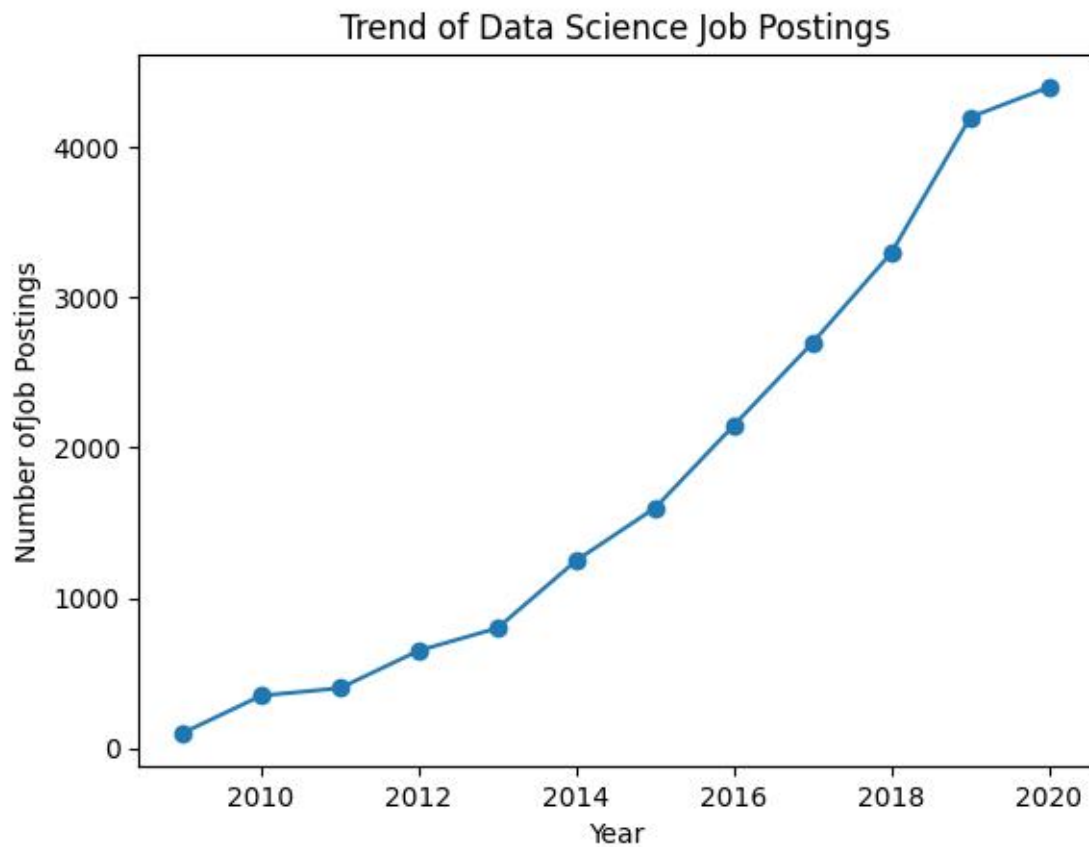
```
plt.title('Trend of Data Science Job Postings')
```

```
plt.xlabel('Year')
```

```
plt.ylabel('Number of Job Postings')
```

```
plt.show()
```

output:



Result:

The graph shows that data science job postings increased from 2009 to 2021. The highest number of job postings was in 2021.

Exp No:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Aim:

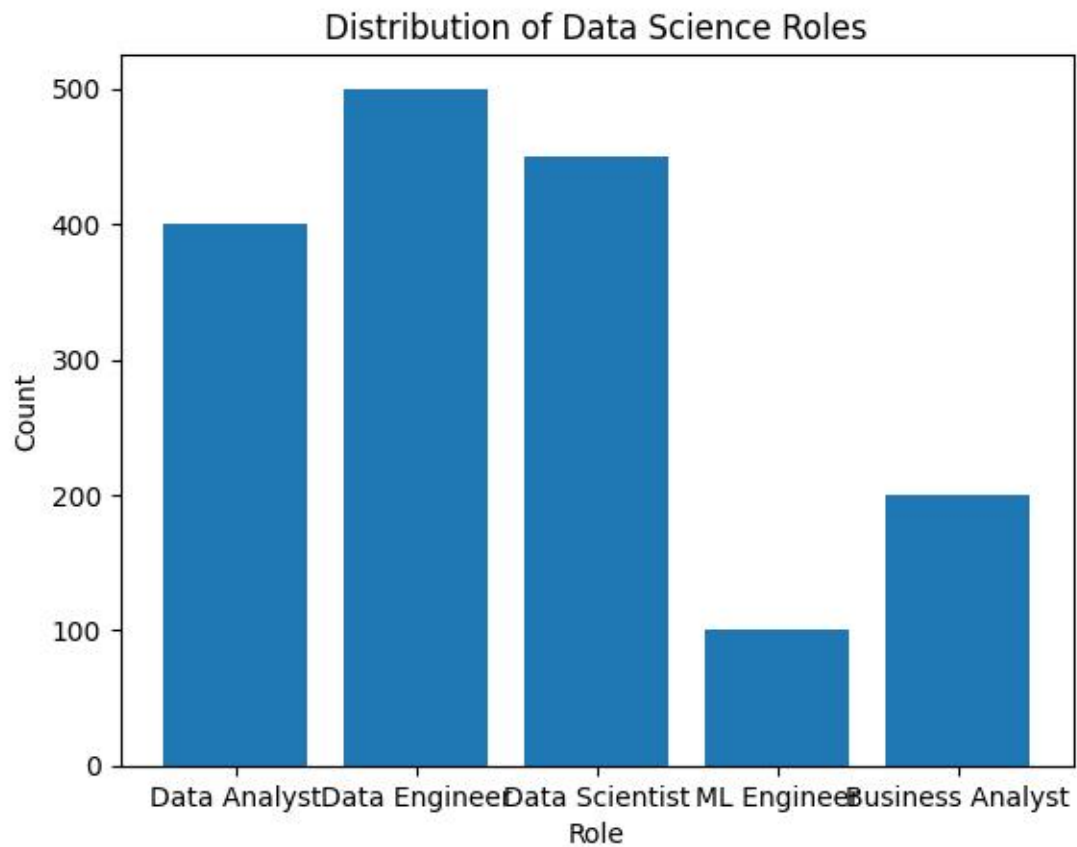
To Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description:

Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

Code:

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',  
'Business Analyst']  
counts = [400, 500, 450, 100, 200]  
plt.bar(roles, counts)  
plt.title('Distribution of Data Science Roles')  
plt.xlabel('Role')  
plt.ylabel('Count')  
plt.show()
```



Result:

Data scientist is occurring rate in the dataset (count = 450). Data analyst (400), Data Engineer (500), and ML Engineer (100) are equally data set.

Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi

structured data based on data sets given.

Aim:

To conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description:

Create small datasets for each type and explain their characteristics.

Code:

```
structured_data = pd.DataFrame({  
'ID': [1, 2, 3],  
'Name': ['Alice', 'Bob', 'Charlie'],  
'Age': [35, 30, 25]  
})  
  
print("Structured Data:\n", structured_data)  
  
unstructured_data = "This is an example of unstructured data. It can be a piece of  
text, an image, or a video file."  
  
print("\nUnstructured Data:\n", unstructured_data)  
  
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}  
  
print("\nSemi-structured Data:\n", semi_structured_data)
```

Output:

Structured Data:

	ID	Name	Age
0	1	Alice	35
1	2	Bob	30
2	3	Charlie	25

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an image, or a video file.

Semi-structured Data:

```
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
```

Result:

The experiment was successfully performed by creating and displaying examples of structured ,unstructured,and semi structured data using python.The characteristics and representation of each type of data were observed successfully.

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

Aim:

To conduct an experiment to encrypt and decrypt given sensitive data.

Description:

Use the cryptography library to encrypt and decrypt a piece of data.

Code:

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Chanakya Matric School")
token
b'...'
f.decrypt(token)
b'Chanakya Matric School'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Chanakya Matric School."
cipher_text = cipher_suite.encrypt(plain_text)
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

output:

Original Data: b'Chanakya Matric School.'

Encrypted Data:

b'gAAAAABqfE_wnADhlHXN2HlerPEuN0B2LwaZ6oHp46s1bxFi1yOIAMcLpVniplxKNr4
ZPxKXHxkmlKfTzAtuU7yrcmm15jj7L2qSBSK3ZS1KmtOYf_HXLfA='

Decrypted Data: b'Chanakya Matric School.'

Result:

The fernet symmetric key algorithm successfully encrypted back original data text ,verifying both any tumering with the cipherset would cause decryption to fail.